

A3-PS004-TC136

Authenticated Gateway-Assisted DNSSEC for IoT Clients

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AIORI Hackathon 3

October 12, 2026

Problem Description

IoT devices need DNSSEC protection but cannot afford to validate it themselves.

- DNSSEC stops spoofing, but checking signatures costs memory, CPU and battery a microcontroller lacks. Without it, a device looking up its OTA or MQTT server can be redirected.
- PS-004 idea: the device hands validation to a trusted local gateway over an authenticated, encrypted channel (e.g. TLS or DTLS).
- Safe only if the device rejects bogus answers *and* refuses unauthenticated gateways; the AD bit alone is just a header flag.
- Goal: measure the device-side cost (latency, CPU, memory) against a baseline, for secure, insecure and bogus answers.
- The device does not validate itself, so protection ends at the gateway; no end-to-end DNSSEC claim.

Background

Three pieces we build on

- **DNSSEC validation** [1, 2]: a validator follows DS/DNSKEY/RRSIG from a trust anchor and labels each answer secure, insecure (no DS, legitimately unsigned) or bogus (signatures fail). Bogus becomes SERVFAIL. Per RFC 6840 [3], a stub may trust the AD bit only over a channel it trusts.
- **Telling the client why:** Extended DNS Errors [4] attach a reason to SERVFAIL, e.g. 6 *DNSSEC Bogus*, 7 *Signature Expired*, 9 *DNSKEY Missing*.
- **Securing the last hop:** DNS over TLS [5] on port 853, or DNS over CoAP [6] (RFC, March 2026) protected by DTLS or OSCORE [7], which encrypts each CoAP message without a full handshake.

Literature & RFC Review

What exists, and what it leaves open

Work	Relevance to us
RFC 4033 / 4035 [1, 2]	Validation states and the AD bit; the reference this problem cites
RFC 6840 [3]	AD bit is only meaningful over a secure path to a trusted validator
RFC 8914 [4]	Extended DNS Errors: the device can log *why* a lookup failed
RFC 7858 / 9325 [5, 9]	DNS over TLS and current TLS settings for that path
RFC 9953 / 8613 [6, 7]	DNS over CoAP with OSCORE: encrypted DNS sized for constrained nodes
Lenders et al., CoNEXT 2023 [8]	DoC vs. DNS over UDP and DTLS on RIOT; OSCORE saves over 10 kB of code vs. DTLS. **No DNSSEC validation or bogus-answer testing.**

Gap: we found no work that measures a delegated validator over both TLS and OSCORE while checking secure, insecure and bogus outcomes on the device.

Proposed Solution

Let a trusted gateway validate, and make the device strict about what it accepts from it

- **Validating gateway:** checks every answer against a trust anchor for our own test domain [10], signed in secure, insecure and bogus variants.
- **Authenticated channel, two ways:** DNS over TLS with mutual authentication, or DNS over CoAP with OSCORE. The gateway identity is fixed on the device, so an impostor is refused.
- **Lightweight device:** ESP32-S3 with no DNSSEC code; trusts only verdicts from the authenticated channel and fails closed without it.
- **Actionable verdicts:** failures carry an Extended DNS Error code, so the device knows *why* a lookup was refused.
- **Critical names must be secure:** an unsigned answer for the update or MQTT server is refused (no silent downgrade).
- **Evaluation:** both channels vs. plain DNS and on-device validation, under spoofing and rogue-gateway attacks.

Optimization Proposed

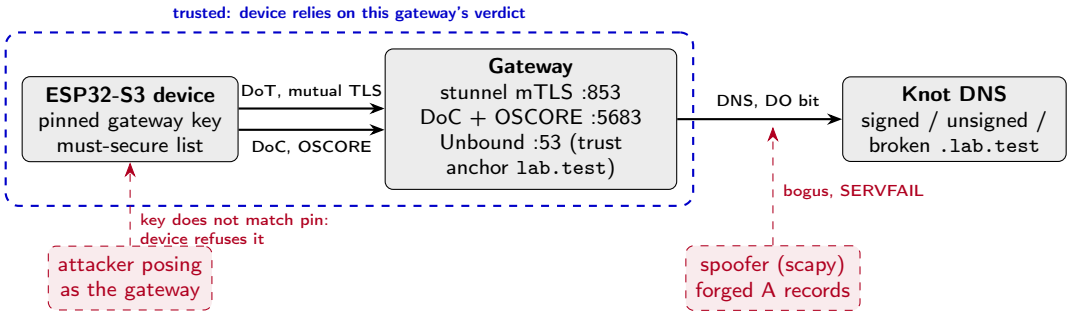
Before vs. after, per lookup on the device

	Plain DNS	**On-device validation**	**Gateway + DoT**	**Gateway + DoC**
Spoofed answer	accepted	rejected	rejected	rejected
Fake resolver / gateway	accepted	forged data rejected	refused (mTLS)	refused (OSCORE)
Unsigned answer, critical name	accepted	accepted	rejected	rejected
Flash / RAM	smallest	largest, may not fit	TLS stack	expected below DTLS [8]
CPU, latency, bytes on air	measured: 30+ runs per case, median and p95			
Energy	only if we get a power monitor (INA219 or similar)			

Only the first three rows are design properties; the rest are what we will measure.

Solution Architecture & Design

Lab setup and the trust boundary



Trust ends at the gateway: the device does not validate DNSSEC itself, so we make no end-to-end claim. Out of scope: a compromised gateway or physical access to the device; both would defeat the scheme. *Figure: System architecture.*

Data Flow Diagram

One lookup, from the device's point of view

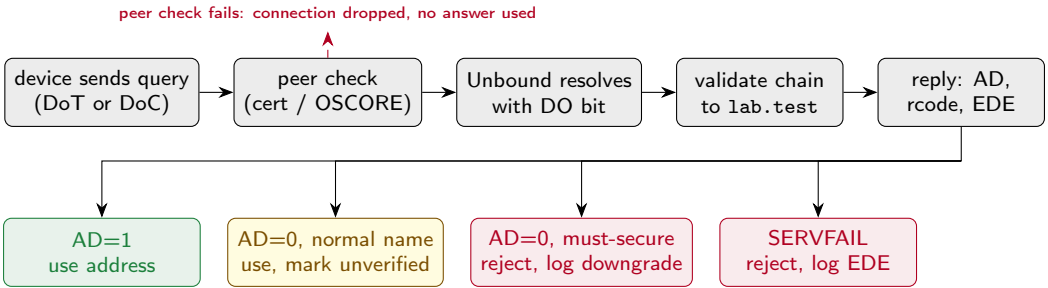


Figure: Data Flow Diagram (DFD)

Deployment Timeline

Plan against the hackathon dates

2026	Work
by 14 Oct	Repo, this deck, team video
16–20 Oct	ESP-IDF toolchain on the ESP32-S3; Knot zones signed; Unbound validating; all three zones checked with `dig +dnssec`
21–25 Oct	stunnel mTLS and DoC/OSCORE proxy; device firmware for both paths; plain-DNS baseline
26–31 Oct	Spoofer and rogue-gateway tests; benchmarks; try on-device validation as the second baseline; code pushed
31 Oct–4 Nov	Report with threat model; recorded demo of the three attacks being rejected
24–30 Nov	Regional round, if selected

Fallback: if DoC/OSCORE slips, the DoT path alone still covers the required results.

Bibliography

[1] R. Arends et al. *DNS Security Introduction and Requirements*. RFC 4033, 2005.

[2] R. Arends et al. *Protocol Modifications for the DNS Security Extensions*. RFC 4035, 2005.

[3] S. Weiler, D. Blacka. *Clarifications and Implementation Notes for DNS Security (DNSSEC)*. RFC 6840, 2013.

[4] W. Kumari et al. *Extended DNS Errors*. RFC 8914, 2020.

[5] Z. Hu et al. *Specification for DNS over Transport Layer Security (TLS)*. RFC 7858, 2016.

[6] M. S. Lenders et al. *DNS over CoAP (DoC)*. RFC 9953, 2026.

[7] G. Selander et al. *Object Security for Constrained RESTful Environments (OSCORE)*. RFC 8613, 2019.

[8] M. S. Lenders et al. Securing Name Resolution in the IoT: DNS over CoAP. *Proc. ACM Netw.* 1(CoNEXT2), 2023. doi:10.1145/3609423

[9] Y. Sheffer, P. Saint-Andre, T. Fossati. *Recommendations for Secure Use of TLS and DTLS*. RFC 9325, 2022.

[10] S. Cheshire, M. Krochmal. *Special-Use Domain Names*. RFC 6761, 2013.

Thank You for Your Attention!